

The Art of Queueing

Reinforcement Learning for visitor flow at the Gallerie degli Uffizi in Florence

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Reinforcement Learning

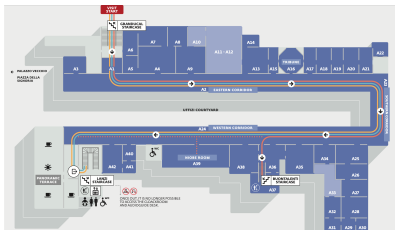
June 2026

The Uffizi Gallery, Florence

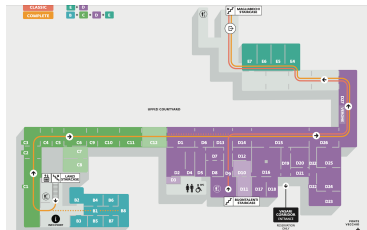
About 5,000 visitors a day move through ninety-eight rooms on two floors. These are the paintings nearly all of them come to see.



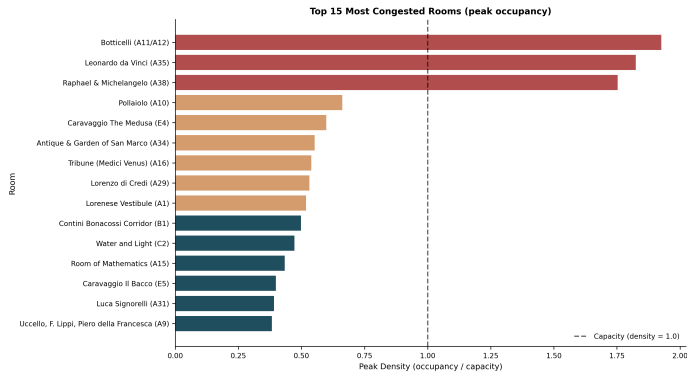
Second floor (the masterpiece rooms)



First floor (Caravaggio, free-flow)



The problem: where RL comes in



At the daily peak the three masterpiece rooms, Botticelli, Leonardo and Raphael, run at roughly twice capacity (red, past the dashed line) while the rest of the museum sits well below. The question: can reinforcement learning help us understand this crowding and simulate a calmer, better-run Uffizi?

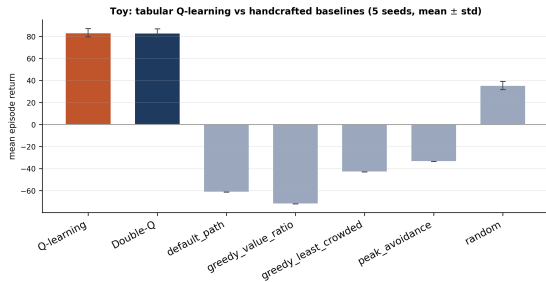
- ① Recreate the Uffizi as a virtual model: every room, its size and doorways, plus a value for each (a Botticelli room is worth far more than a stairwell).
- ② Generate the crowd, today's arrivals and mix, about 60% phone-first tourists, 30% ordinary visitors, 10% art lovers. Validate it against the density maps (does it look like the Uffizi today?), then read off today's revenue and average welfare.
- ③ Drop an RL agent on the museum as-is: the optimal visitor today.
- ④ Intervene: change hours, room access and pricing; re-simulate; keep only changes that hold or raise revenue while lifting welfare, because a revenue-losing redesign is never adopted.
- ⑤ Re-run the RL agents on the intervened museum: did the optimal visit improve for every kind of visitor?

A toy first: learning beats fixed rules

We prove it on a 12-room toy with an exact value table. Crowds wave up and down, unannounced.

Tabular Q-learning learns temporal arbitrage: enter in the lull around the crowd peak.

Across five seeds: Q and Double-Q tie (83.3 vs 83.1). A deterministic toy has no overestimation bias to remove, which is just what the theory says.



Learning beats every hand-written rule; the rules even go negative by mistiming the crowd.

The toy does not scale: deep RL and masking

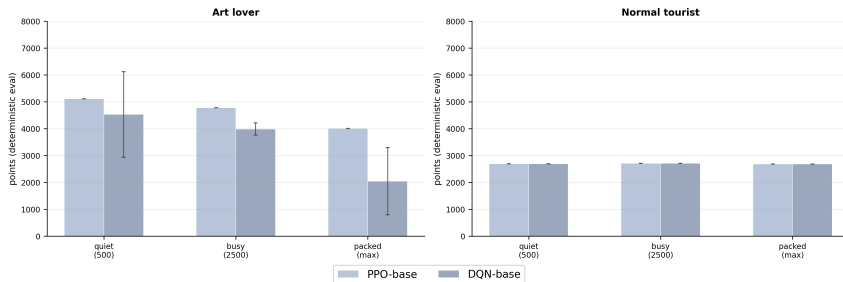
The real museum breaks the table: ninety-eight rooms, ~ 100 moves per step, hundreds of state numbers. The same Q-table could never fill, so we change algorithm, to a neural net (MaskablePPO; PPO and DQN as controls).

Action masking: the rulebook

From any room only a few moves are legal, yet the net has ~ 100 outputs. We set every illegal move to $-\infty$ before the softmax, so it gets exactly zero probability. So no gradient goes into the moves that cannot happen; the agent only ever picks among the legal ones.

The museum as it is: the baseline

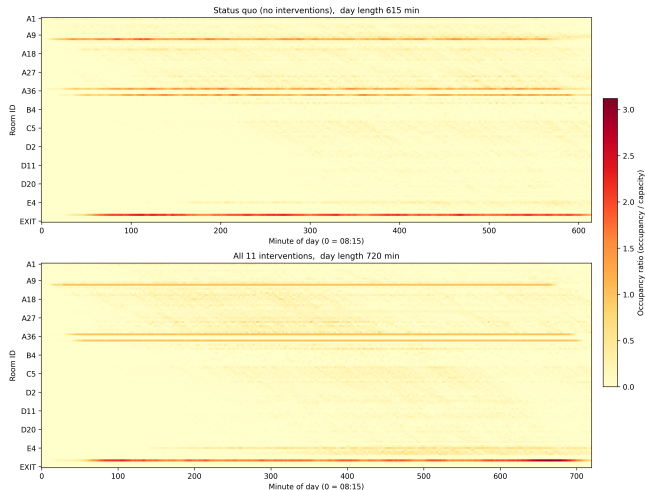
The museum as it is: the optimal visitor with no intervention (5 seeds, mean $\pm$ std)



With the virtual museum calibrated to today (real hours, prices and surcharges), we drop the RL agents on the museum as it is, where the visitor just walks. This is the ceiling with no intervention: the art lover slips from $\sim 5,100$ to $\sim 4,000$ as the rooms fill; the tourist holds near $\sim 2,700$. That is today's Uffizi, the baseline we compare everything against.

The interventions and the museum-wide payoff

Crowd density heatmap: status quo vs all 11 interventions (~5000 visitors). Shared colour scale.



Eleven Pigovian measures, each pricing the crowding a visitor causes:

- Time: longer hours, quiet-hour discounts.
- Access: reservations (RAMA), enrich the side rooms.
- Price: peak pricing, group surcharge.

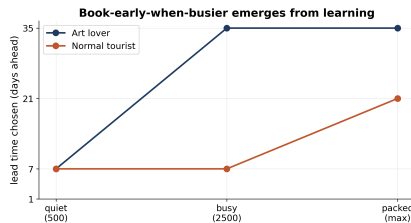
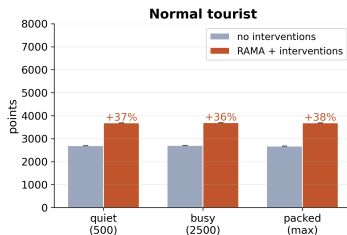
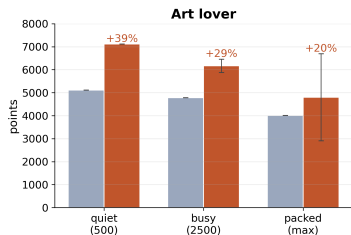
Top: as-is; bottom: all eleven, same scale. The masterpiece bands cool from red toward orange; the day spreads over more hours.

The director's test is met: **+31%** welfare, revenue held.

After the redesign: the visit becomes a booking problem

RAMA gates the masterpieces behind booked slots, so a visitor can no longer just walk in. The real decision flips from where to walk to when to book (lead time 1, 7, 21 or 35 days), hard because it is uncertain, irreversible and delayed.

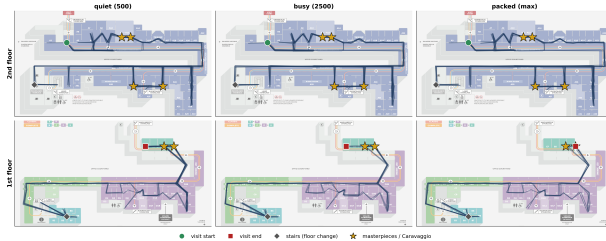
RAMA booking: intervened vs matched baseline (5 seeds, mean $\pm$ std; 3/3 masterpieces secured)



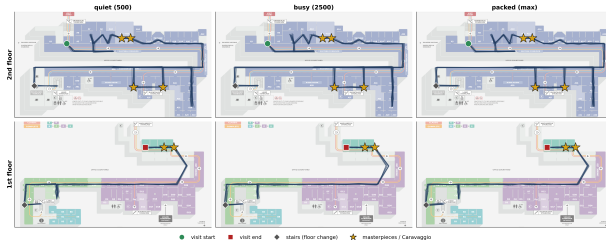
The ceiling rises at every crowd level. The art lover's baseline $\sim 5,100/4,000$ climbs to $\sim 7,100/4,800$ (+20 to +39%); the tourist's $\sim 2,700$ climbs to $\sim 3,690$ (+36 to +38%), all three masterpieces secured. We never coded book-early-when-busier (right); it came straight out of the reward.

Two visits: the art lover roams, the tourist beelines

Art lover walking routes (RAMA + interventions) -- one continuous visit, 8 rollouts/panel; It descends floor 2 -> floor 1.



Normal tourist walking routes (RAMA + interventions) -- one continuous visit, 8 rollouts/panel; It descends floor 2 -> floor 1.



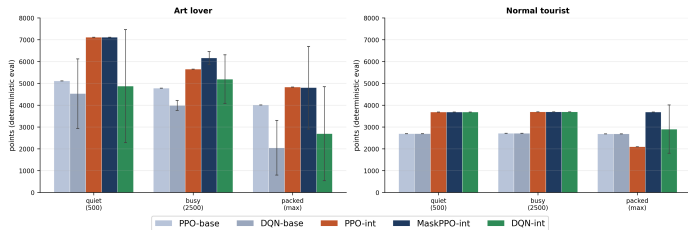
Top: the art lover. It works the whole museum: the trace covers the second floor end to end and fans across the first-floor blocks too.

Bottom: the normal tourist. It does not. After the masterpieces it skips the B block and hurries through C and D straight to Caravaggio, then leaves.

Same museum, two very different visits. (Green = start, red = exit; stars mark the masterpieces and Caravaggio.)

Which algorithm wins and why masking matters

Algorithm comparison: value-based (DQN) vs policy-gradient (PPO / MaskablePPO), baseline vs intervened (5 seeds, mean $\pm$ std)



- Pale bars: the two no-booking baselines. Coloured: the three booking agents.
- MaskablePPO (dark blue) is best or tied in every cell and stable across seeds: the winner.
- Unmasked PPO (orange) **collapses** at the packed tourist (2093, below the 2680 baseline): it wastes effort on illegal moves. Masking only matters at the hard cells, but there it decides the result.
- DQN (green) is crowd-fragile and high-variance (the wide error bars): a max over noisy crowd values overestimates, the bias the deterministic toy did not have.

1: an unlearnable reward

S Symptom. The agent never left, squatting in one room until closing. Zeroing all reward for not exiting made it worse.

D Diagnosis. An all-or-nothing wipe is flat almost everywhere, so the gradient is zero: nothing points to the door.

F Fix. Use a dense egress signal: per-step pressure rising near closing and far from an exit, plus a bonus for leaving.

L Lesson. Even a correct reward is useless if it is flat everywhere. Gradient descent needs a slope to follow, so we built one in.

2: it could not see what it was deciding

S Symptom. The rule “stay until satisfied, then move” never stabilised; dwell times wobbled at random.

D Diagnosis. The right action depended on value already taken from the room, not in the state: a hidden variable the agent could not see.

F Fix. Add a per-room appreciation-progress value to the state, so “seen enough?” becomes something the policy can read.

L Lesson. Whatever the best move depends on has to live in the state. We had left it out, so no policy could settle.

3: reward shaping that backfired

S Symptom. Well-meant terms backfired: a “boredom” penalty caused looping, a harsh crowd penalty made a fake art lover.

D Diagnosis. Each reward term we added gave the agent a new way to cheat: looping beat exiting, a crowded Botticelli scored worse than just skipping it.

F Fix. Minimal shaping: a gentle crowd discount so a crowded masterpiece still beats skipping, no boredom penalty, satiation alone stops lingering.

L Lesson. The agent games every term the laziest way it can. We stopped adding shaping until we had checked what each term really paid for.

4: the do-nothing trap and the book-early trick

S Symptom. Given a free “decline” button, both profiles declined all three masterpieces; book-early stayed only half learned.

D Diagnosis. Declining is safe now, booking pays only later: a do-nothing optimum; its reward lands hundreds of steps away (credit assignment).

F Fix. Make “always books” a type trait with no escape hatch; add an immediate off-pace penalty proxying the distant cost.

L Lesson. A small penalty now, standing in for a cost hundreds of steps away, was what finally got it booking. Nothing else we tried helped as much.

Conclusion

We rebuilt the Uffizi, matched it to the real museum, then asked two questions: how to redesign the building and how a single visitor should play it.

- The redesign: eleven Pigovian measures lift welfare **+31%** with revenue held.
- The visitor: the optimal agent (MaskablePPO) books early and beats its as-is ceiling at every crowd level.
- The lesson: choosing what the agent decides was far harder than the algorithm itself; action masking did more for us than any amount of tuning.

Thank you.

Questions?

Appendix

Backup slides for questions

Backup: the three MDPs at a glance

Setting	State	Action	Reward	γ
Toy (Q, Double-Q)	5-tuple table	stay / move	toy per-step	0.99
As-is (PPO, DQN)	8 numbers	stay / move on	walk, no booking	0.995/0.997
RAMA (MaskablePPO, PPO, DQN)	25 numbers	lead / book / pace masked for MPPO	booking per-step	0.999

The same five elements in every setting; what changes is the problem the agent is attached to, plus how the state is stored and the actions screened. PPO and DQN appear twice, once as a no-booking walk (the baselines) and once as booking agents; MaskablePPO only on RAMA.

Backup: the as-is state, eight numbers

The visitor's local view on the planned walk (natural units; the network sees each scaled to about $[0, 1]$):

- 1 Room-value drain: minutes absorbed vs the room's worth (~ 3 min per importance point).
- 2 Room value: importance, 1 to 10.
- 3 Density: how full the room is, percent of capacity.
- 4 On-target flag: are you in the next planned must-see?
- 5 Plan completed: share of must-sees finished.
- 6 Time of day: minute of the 615-minute day.
- 7 Egress slack: spare minutes to reach an exit before closing.
- 8 Distance to next room: doorways to the next must-see.

At the door A1: (0, 0.22, 0.30, 0, 0, 0, 0.99, 0.10).

Backup: the RAMA state, twenty-five numbers

A two-phase episode (a booking screen, then the walk), grouped:

- Booking position (3): phase flag, which of 4 decisions, lead chosen (days).
- Booking screen (4): slots free at lead 1/7/21/35 days (out of 8).
- Time of day (1): minute of the 720-minute day.
- Windows held (3): booked clock-times for Botticelli, Leonardo, Raphael.
- Check-ins (3): which masterpieces collected.
- Access read (4): is-masterpiece, unbooked, open-now, minutes-to-open.
- Pace prior (3): when it expects to reach each masterpiece (frozen).
- Room tail (4): drain, value, density, egress, as in the as-is state.

Opening screen: (1, 0, 1; 0.12, 0.38, 0.62, 1; 0; ...); the four slot counts are what the booking decision reads.

Actions per setting

- Toy / as-is: Discrete(2), stay or move on.
- RAMA: Discrete(4). Booking phase: a lead $\in \{1, 7, 21, 35\}$ or book/decline.
Walk phase: stay or move on.

Masking (MaskablePPO). Set the logits of illegal moves to $-\infty$ before the softmax, so they get exactly zero probability:

$$\pi(a \mid s) = \frac{e^{z_a} \mathbf{1}[a \in \mathcal{A}(s)]}{\sum_b e^{z_b} \mathbf{1}[b \in \mathcal{A}(s)]}.$$

Backup: the reward functions

Toy, per step. $\frac{\text{importance} \times \text{novelty}}{1 + \alpha_A \text{density}^2} - \text{step cost}$, with $\alpha_A = 6$, step 0.5, exit bonus +15, no-exit penalty -50 (hand-tuned shaping; only the relative sizes matter).

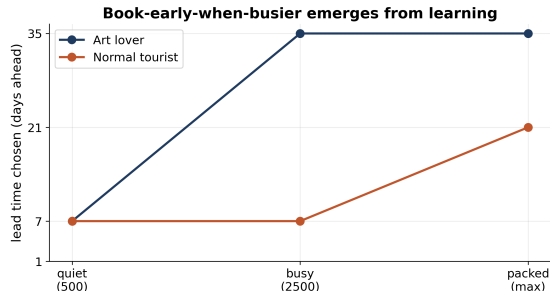
Full museum, per minute. $\text{importance} \times \text{crowd discount} \times \text{novelty}$ (satiation), with a gentle crowd $\alpha = 0.5$, plus closing-time pressure, a completion bonus and a small step cost.

Booking terms (RAMA only). an immediate off-pace penalty at commit time (a local proxy for the distant cost of a badly-timed slot) plus a no-show penalty for a masterpiece booked but missed.

Two-phase episode

- Booking (no time passes): pick a lead time (1/7/21/35 days), then book each masterpiece.
- Windows committed near the expected arrival; then the minute-by-minute walk.

Fill curve (busy day). Slots free out of 8: 1 a day before, 3 a week ahead, 5 three weeks ahead, 8 a month ahead. Book late, miss the masterpiece.



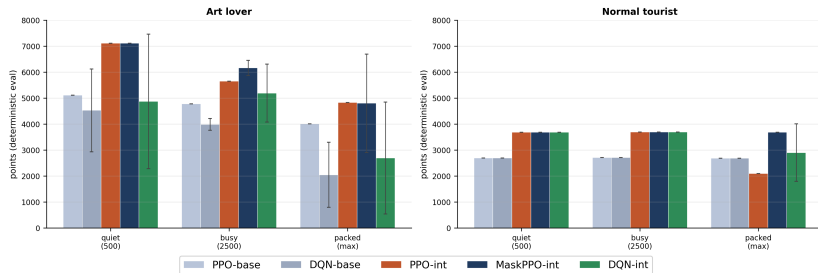
Backup: full booking results, five seeds (mean $\pm$ std)

Profile	Crowd	Baseline	Intervened	Gain	Check-ins
Art	500	5112	7113 $\pm$ 0	+39%	3.0/3
Art	2500	4780	6167 $\pm$ 291	+29%	3.0/3
Art	5000	4010	4800 $\pm$ 1891	+20%	2.6/3
Tourist	500	2694	3686 $\pm$ 0	+37%	3.0/3
Tourist	2500	2708	3695 $\pm$ 0	+36%	3.0/3
Tourist	5000	2680	3685 $\pm$ 0	+38%	3.0/3

Baseline = best no-booking walk (ppo_base); intervened = MaskablePPO on RAMA. Five training seeds {1, 3, 7, 42, 202606}, each evaluated on six fixed common-random-number crowds. The art lover at max crowd is bimodal: four seeds reach $\sim +41\%$, one collapses, hence the wide band and the 2.6/3 check-ins.

Backup: which algorithm and where it breaks

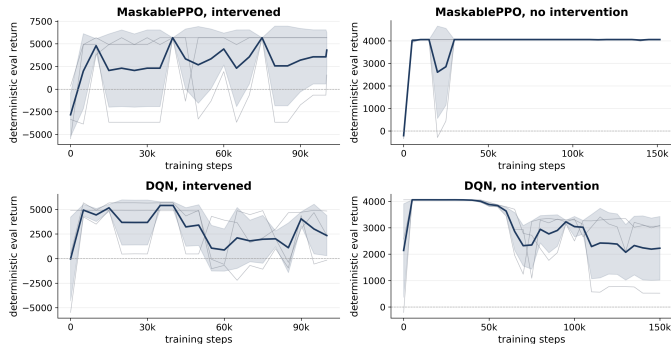
Algorithm comparison: value-based (DQN) vs policy-gradient (PPO / MaskablePPO), baseline vs intervened (5 seeds, mean $\pm$ std)



- MaskablePPO: best or tied in every cell, near-zero seed variance.
- Unmasked PPO: collapses at the packed tourist to 2093, below the 2680 no-booking baseline, while MaskablePPO holds 3685.
- DQN: crowd-fragile and high-variance, its art baseline falls 4527 $\rightarrow$ 2044 as the crowd grows (a max over noisy values overestimates).

Backup: how each algorithm learns over time

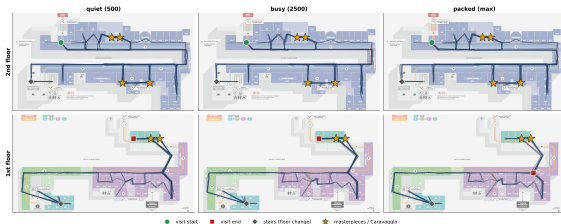
How each algorithm learns: deterministic eval over training (art lover, packed crowd; 3 seeds, mean $\pm$ std)



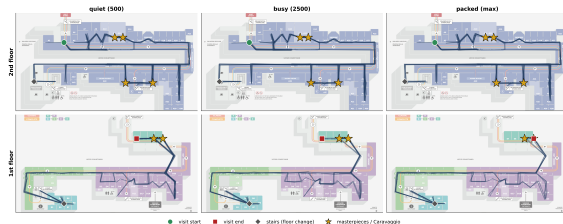
Deterministic eval (greedy, on the fixed crowds) taken every 5k steps, for the art lover at the packed crowd, 3 seeds. MaskablePPO with no intervention reaches its ceiling early and stays there. DQN climbs, then degrades as overestimation builds on the noisy crowd. MaskablePPO intervened is unstable across seeds: most stay high, one collapses.

Backup: the learned walk, art lover

Art lover walking routes (no interventions) – one continuous visit, 8 rollouts/panel; it descends floor 2 -> floor 1.



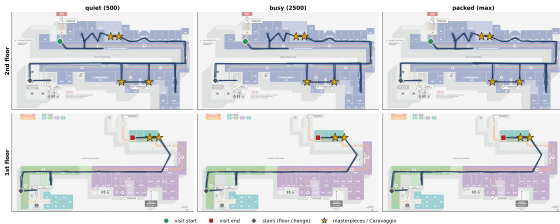
Art lover walking routes (RAMA + interventions) – one continuous visit, 8 rollouts/panel; it descends floor 2 -> floor 1.



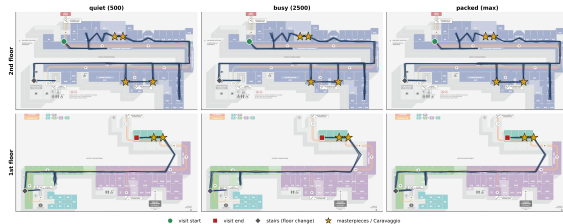
Left: as-is. Right: under RAMA. Eight rollouts per panel across the three crowd levels; the masterpieces (gold stars) are taken in their booked windows, then a descent to Caravaggio and out.

Backup: the learned walk, tourist

Normal tourist walking routes (no interventions) -- one continuous visit, 8 rollouts/panel; it descends floor 2 -> floor 1.



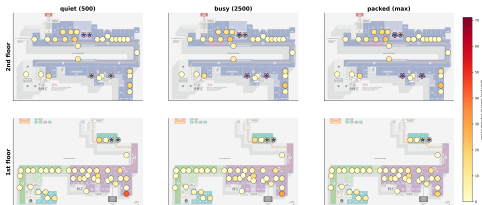
Normal tourist walking routes (RAMA + interventions) -- one continuous visit, 8 rollouts/panel; it descends floor 2 -> floor 1.



Left: as-is. Right: under RAMA. The tourist books all three masterpieces and stays stable across seeds at every crowd level.

Backup: per-profile crowd density

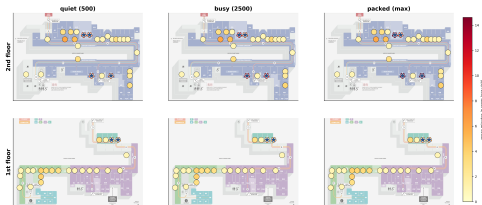
Art lover – where time is spent (no interventions). Darker = more minutes. Blue stars = masterpieces/Caravaggio.



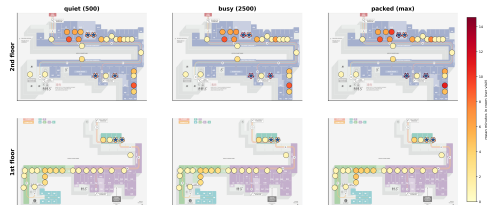
Art lover – where time is spent (RAMA + interventions). Darker = more minutes. Blue stars = masterpieces/Caravaggio.



Normal tourist – where time is spent (no interventions). Darker = more minutes. Blue stars = masterpieces/Caravaggio.

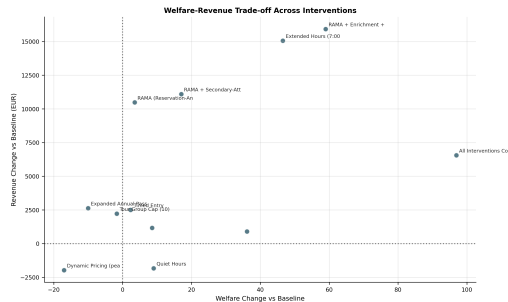


Normal tourist – where time is spent (RAMA + interventions). Darker = more minutes. Blue stars = masterpieces/Caravaggio.

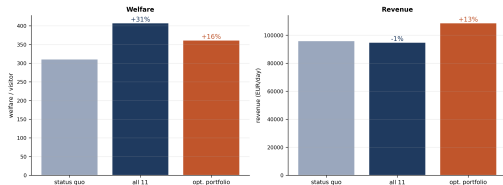


Left column as-is, right column under RAMA; art lover on top, tourist below. The booked windows spread each profile's masterpiece visits across the day.

Backup: the interventions and the director's gate

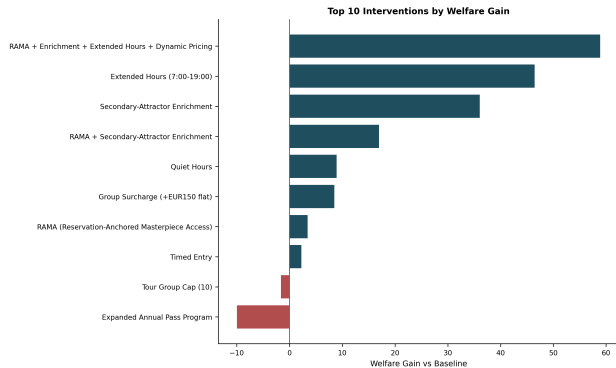


Museum-wide director gate (as-is crowd): all 11 lift welfare with revenue preserved; the optimized subset raises both



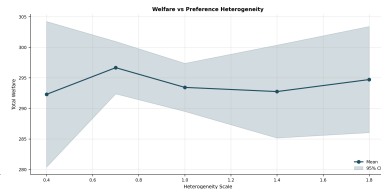
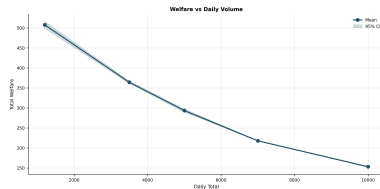
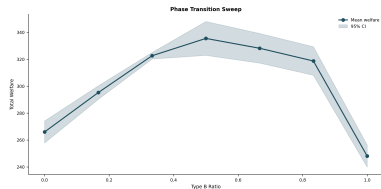
The redesign is adopted only if welfare rises and revenue does not fall (the director's gate). The chosen portfolio of eleven Pigovian measures lifts welfare **+31%** with revenue preserved.

Backup: ranking the candidate interventions by welfare gain



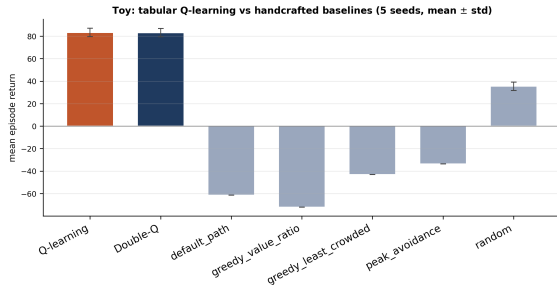
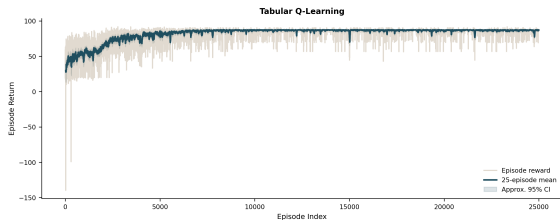
Each candidate measure scored on its own against the as-is baseline. Extended opening hours and secondary-attractor enrichment are the strongest single levers; the full combined bundle leads. The tour-group cap and the expanded annual pass lower welfare on their own; the adopted portfolio keeps the welfare-positive measures.

Backup: robustness sweeps



We vary the Type-B (Instagram) share, the total daily volume and the crowd heterogeneity. The welfare gain from the redesign holds across the ranges we tried.

Backup: tabular toy internals



Left: the learning curve, return climbing over episodes. Right: learning beats every hand-written rule. Across five seeds Q (83.3 ± 3.8) and Double-Q (83.1 ± 3.7) tie, since a deterministic toy has no overestimation bias for Double-Q to remove.

Backup: hyperparameters, every arm

Shared by every PPO-family arm (Stable-Baselines3): MlpPolicy [128,128], Adam, $n_{\text{steps}} = 2048$, batch 256, 10 epochs, GAE $\lambda = 0.98$, clip 0.2, entropy 0.01.

Arm	Net	LR	γ	Budget
Toy Q / Double-Q	tabular	$\alpha = 0.1$	0.99	25k episodes
PPO baseline (walk)	[128, 128]	3×10^{-4}	0.995/0.997	150k steps
MaskablePPO (RAMA booking)	[128, 128]	3×10^{-4}	0.999	100k steps
PPO intervened (no mask)	[128, 128]	3×10^{-4}	0.999	100k steps
DQN (baseline / intervened)	[128, 128]	5×10^{-4}	0.995–0.997 / 0.999	150k / 100k

Toy Q: ε from 1.0 to a small floor, $\times 0.9995$ each episode. **DQN extras:** replay buffer 100k, target update every 1000, train-freq 4, learning-starts 2k, ε 0.3 $\rightarrow$ 0.05. **MaskablePPO** adds the legal-move mask; the booking arms train directly at each crowd, with no curriculum. **Scale:** one step = one minute, so a visit is the 720-minute day plus 4 booking decisions, about 720 steps; 100k steps $\approx$ 140 visits; a 2048-step rollout $\approx$ 3.

Backup: what each setting means

- **Network / learning rate:** two hidden layers of 128 units; how big a step the weights take on each update.
- n_{steps} / **batch** / **epochs:** steps collected before learning (2048); the chunk size per weight step (256); how many times that batch is reused, one full pass = one epoch (10).
- γ (**discount**): how far ahead reward counts; 0.999 keeps a payoff hundreds of steps away alive, which booking needs.
- **GAE** $\lambda = 0.98$: grade a move by what happened later ($\lambda \rightarrow 1$) or the critic's one-step guess ($\lambda \rightarrow 0$); 0.98 leans on what really happened.
- **clip** 0.2: PPO's brake; one update can't shift an action's probability past about 20%.
- **entropy** 0.01: entropy is how undecided the policy is; a small bonus for staying spread out, so it explores before committing.
- **DQN replay / target** / ϵ -**greedy**: reuse past experience, chase a slow target, explore early then exploit.

Seeds and evaluation: 5 training seeds {1, 3, 7, 42, 202606}; deterministic eval on 6 fixed common-random-number crowds (900000–900005); crowd levels 500/2500/5000 a day.

Backup: limitations and where we would go next

One visitor, not a crowd. What we optimise is a single sophisticated visitor's best response to a fixed crowd. It is not an equilibrium: if every visitor booked this way the good slots would fill faster and the fill curve itself would move. We did not model that population response, which would be a mean-field game.

Fixes we did not run. Our toy isolates the overestimation that makes DQN crowd-fragile, yet we never tried Double-DQN, its standard remedy, on the full task. We also did not add a fixed-heuristic booker ("always reserve a month out") as a tougher baseline than no-booking, nor an ablation turning the off-pace penalty off to test whether booking early is learned or only forced by that penalty.

What is baked in. The connoisseur always books (a fixed trait, not a learned choice); lead times are four discrete values; the fill curve is calibrated rather than fitted to real booking data; the policy is a plain feedforward net carrying appreciation-progress, with $\gamma = 0.999$. Welfare is the simulator's own visitor-utility, itself a modelling choice and not a settled social objective.